Devixo Solutions

AI/ML Internship Program

Task 02 Build and Compare Machine Learning Models

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1. Introduction

This project develops and compares regression models to predict a student Performance Index from study habits, previous scores, sleep hours, extracurricular participation, and practice-question activity. The work follows a machine-learning workflow: inspect and clean the data, prepare features, train multiple models, evaluate their test-set performance, and tune tree-based models using GridSearchCV.

2. Objective

- Prepare the dataset for regression modelling.
- Train and compare Linear Regression, Decision Tree, and Random Forest models.
- Evaluate models using MAE, MSE, RMSE, and R^2.
- Use GridSearchCV to tune Decision Tree and Random Forest hyperparameters.
- Select a recommended model based on measured test-set performance and training time.

3. Dataset Name and Source

Kaggle dataset: Student Performance (Multiple Linear Regression)

URL: https://www.kaggle.com/datasets/nikhil7280/student-performance-multiple-linear-regression

4. Dataset Description

The dataset contains student-related input variables and a numerical target called Performance Index. The task is a supervised regression problem because the target is numeric. The five predictor columns describe hours studied, previous scores, extracurricular participation, sleep hours, and the number of sample question papers practised.

5. Dataset Size

Stage	Rows	Columns
Original dataset	10,000	6
After duplicate removal	9,873	6
Training set	7,898	5 input features
Test set	1,975	5 input features

6. Features/Columns Description

Column	Description
Hours Studied	Number of hours a student studied.
Previous Scores	Student's previous score value.
Extracurricular Activities	Whether the student participates in extracurricular activities; encoded as Yes = 1 and No = 0.
Sleep Hours	Reported hours of sleep.
Sample Question Papers Practiced	Number of sample question papers practised.
Performance Index	Numeric target variable predicted by the regression models.

7. Methodology

1. Load the CSV dataset and inspect its shape, column names, data types, missing values, and duplicates.
2. Remove duplicate records and encode the categorical extracurricular-activities column.
3. Separate the five input features (X) from the Performance Index target (y).

4. Split the data into 80% training and 20% testing sets using random_state = 42.
5. Fit StandardScaler on the training features and use it to transform the training and test features.
6. Train Linear Regression, Decision Tree Regressor, and Random Forest Regressor.
7. Measure training time and calculate MAE, MSE, RMSE, and R^2 on the test set.
8. Use 5-fold GridSearchCV with negative mean squared error scoring to tune Decision Tree and Random Forest.
9. Compare all five configurations and visualize the evaluation results.

8. Data Loading

The dataset was loaded into a pandas DataFrame in Google Colab. Initial inspection showed 10,000 rows and 6 columns. The project used pandas and NumPy for data handling, scikit-learn for preprocessing and modelling, and Matplotlib/Seaborn for visualizations.

9. Data Cleaning

- The initial dataset contained 10,000 rows and 6 columns.
- No missing values were reported during the initial inspection.
- A total of 127 duplicate rows were identified and removed.
- The cleaned dataset contained 9,873 rows and 6 columns.
- The categorical value in Extracurricular Activities was encoded as Yes = 1 and No = 0.
- The scaler was fitted only on the training data to avoid using test-set statistics during preprocessing.

10. Exploratory Data Analysis (EDA)

EDA was used to inspect the dataset structure, review feature types, check missing values and duplicates, and explore relationships among numeric variables.

11. Statistical Analysis:

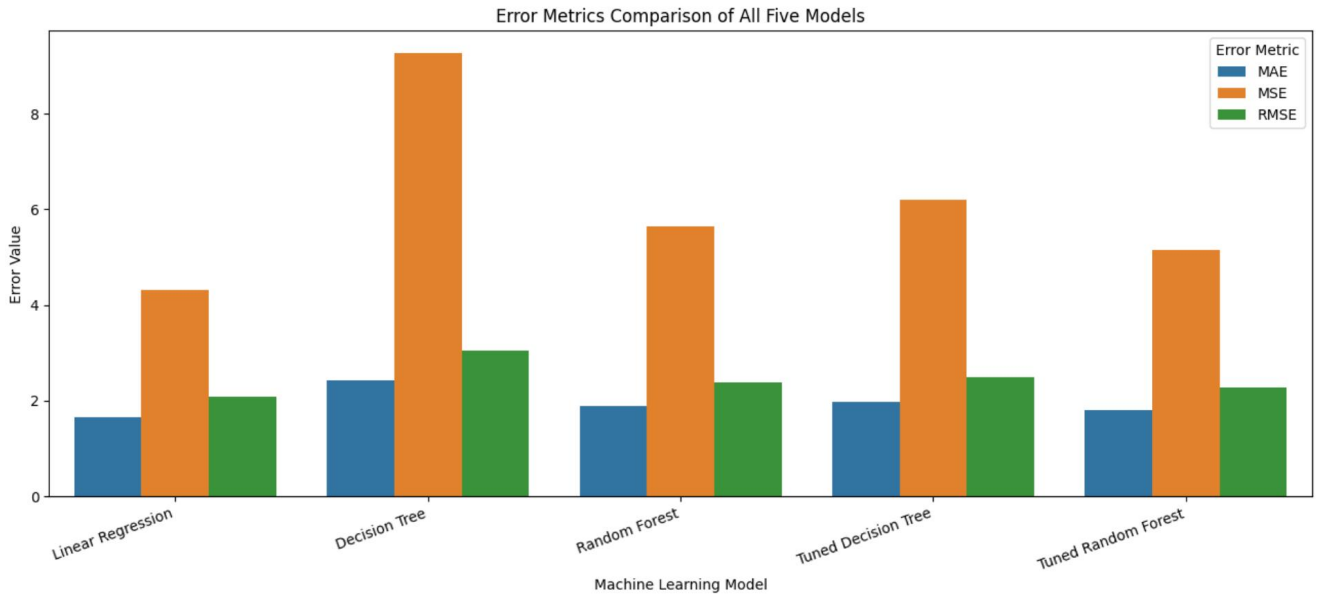
The initial data inspection reported no missing values and 127 duplicate rows. The cleaned dataset contained 9,873 records. The project used a training/test split of 7,898 and 1,975 records respectively.

12. Graph Screenshots

Graph 1 — R^2 Comparison



Graph 2 — Model Error Comparison



13. Major Findings

- Linear Regression achieved the best overall test-set metrics among the five model configurations.
- Linear Regression produced MAE = 1.6470, MSE = 4.3059, RMSE = 2.0751, and R^2 = 0.9884.
- The baseline Decision Tree had the largest errors among the three original models.
- The baseline Random Forest performed better than the baseline Decision Tree but did not outperform Linear Regression.
- GridSearchCV improved both tree-based models on the held-out test set compared with their untuned versions.
- The tuned Random Forest outperformed the tuned Decision Tree, but Linear Regression remained the strongest overall model.

14. 10 Insights

10. The original dataset contained 10,000 rows and 6 columns.
11. The dataset had 127 duplicate rows, which were removed before modelling.
12. After duplicate removal, 9,873 rows remained.
13. The dataset had no missing values according to the initial missing-value check.
14. There were five predictor features and one numeric target column.
15. The data was split into 7,898 training rows and 1,975 test rows.
16. Linear Regression had the lowest MAE among the five evaluated configurations (1.6470).
17. Linear Regression had the lowest MSE among the five evaluated configurations (4.3059).
18. Linear Regression had the lowest RMSE among the five evaluated configurations (2.0751).
19. Linear Regression had the highest R^2 score among the five evaluated configurations (0.9884).
20. The baseline Decision Tree had RMSE = 3.0440, which was higher than the baseline Random Forest RMSE of 2.3732.
21. The tuned Decision Tree RMSE (2.4883) was lower than the baseline Decision Tree RMSE (3.0440).
22. The tuned Random Forest RMSE (2.2713) was lower than the baseline Random Forest RMSE (2.3732).
23. The baseline Linear Regression training time was about 0.0291 seconds, compared with 2.2213 seconds for the baseline Random Forest in the recorded run.
24. The tuned Random Forest improved over the baseline Random Forest on the test metrics but still did not beat Linear Regression.

15. Conclusion

The project compared five regression models to predict student performance. After cleaning and preprocessing the data, Linear Regression achieved the best results (R^2 = 0.9884, MAE = 1.6470). Although tuning improved Decision Tree and Random Forest, neither outperformed Linear Regression, making it the recommended final model.

16. GitHub Repository Link

Repository URL: https://github.com/shafaque-shahid/Devixo_Solutions_Task02